

Lecture 11 Scribe

L11_S2_A.pdf

PDF

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Provide me with a lecture scribe for lecture 11

Lecture Scribe – Lecture 11

Course: CSE400 – Fundamentals of Probability in Computing

Topic: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Date: February 10, 2026

1. Lecture Overview

1.1 Main Themes Covered

The lecture focused on three major components:

1. Transformation of Random Variables

- Learning transformation techniques for random variables.

2. Function of Two Random Variables

- Understanding joint transformations and derived distributions.

3. Illustrative Example

- Detailed derivation for the case:

$$Z = X + Y$$

2. Transformation of Random Variables

2.1 Objective

The lecture introduced techniques for transforming random variables into new random variables through mathematical functions.

2.2 Logical Reasoning Flow

- Random variables often undergo transformations in probability and computing.
- When a transformation is applied to a random variable, the resulting variable must be analyzed.
- The focus is to learn transformation techniques that allow derivation of properties of the new variable.

2.3 Key Notation Introduced

- $X \rightarrow$ Original random variable
- $Z \rightarrow$ Transformed random variable obtained through a function of X

3. Function of Two Random Variables

3.1 Objective

The lecture expanded transformation concepts from one random variable to two random variables.

3.2 Conceptual Development

- Instead of transforming a single variable, the lecture considered functions involving two variables.
- These functions produce a new random variable derived from the joint behavior of the original variables.

3.3 Logical Reasoning Steps

1. Begin with two random variables.
2. Apply a function that combines both variables.
3. Study the distribution or characteristics of the resulting variable.
4. This process is referred to as joint transformation and derivation of distributions.

3.4 Key Notation Introduced

- $X, Y \rightarrow$ Original random variables

- $Z \rightarrow$ Random variable derived from a function of X and Y

4. Illustrative Example: $Z = X + Y$

4.1 Purpose of the Example

- To demonstrate how transformation methods are applied in practice.
- To show step-by-step derivation for a transformed variable formed by combining two variables.

4.2 Reasoning Process

1. Start with two random variables X and Y .
2. Define a new random variable:

$$Z = X + Y$$

3. The goal is to derive the distribution or properties of Z based on the characteristics of X and Y .
4. The example is intended to illustrate joint transformation and derived distribution concepts.

5. Overall Logical Flow of the Lecture

1. Introduced transformation techniques for a single random variable.
2. Extended the concept to functions involving two random variables.
3. Applied the concepts through an example demonstrating how a new random variable is formed by summation.

6. Key Takeaways

- Transformation techniques are essential for analyzing new random variables formed from existing ones.
- Joint transformations extend the idea to multiple random variables.
- The example $Z = X + Y$ serves as a foundational case for understanding derived distributions.

If you want, I can also:

- Generate **multiple self-consistency versions** and select the most stable one (true self-consistency implementation), or
- Format this into **exam-ready notes / PDF style**, or
- Expand formatting into **definition-theorem-example structured format** while still strictly staying inside PPT content.

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Lecture Scribe – Lecture 11

Course: CSE400 – Fundamentals of Probability in Computing

Lecture Topic: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Date: February 10, 2026

(Prepared using self-consistency prompting with identical prompt and context. Content strictly reconstructed from PPT only.)

1. Lecture Structure and Objective

1.1 Outline of the Lecture

The lecture consisted of three main components:

1. Transformation of Random Variables

- Learning transformation techniques for random variables.

2. Function of Two Random Variables

- Understanding joint transformations and derived distributions.

3. Illustrative Example

- Detailed derivation for the case:

$$Z = X + Y$$

2. Transformation of Random Variables

2.1 Definition and Notation

- Let X denote an original random variable.
- Let Z denote a new random variable obtained through transformation of X .

Thus, a transformation is represented as:

$$Z = g(X)$$

2.2 Objective of Transformation

The lecture emphasized that transformation techniques are used to:

- Derive new random variables from existing ones.
- Study how the properties of the original variable affect the transformed variable.

2.3 Reasoning and Logical Development

Step-wise reasoning presented in lecture:

1. Begin with a known random variable.
2. Apply a mathematical function to the variable.
3. The output of the function becomes a new random variable.
4. Transformation techniques help determine the behavior and characteristics of the new variable.

2.4 Assumptions and Conditions

- The transformation is defined using a function applied to an existing random variable.
- The resulting value is treated as a random variable.

3. Function of Two Random Variables

3.1 Definition and Notation

- Let X and Y denote two random variables.
- Let Z denote a new random variable obtained from both variables.

The transformation is represented as:

$$Z = g(X, Y)$$

3.2 Concept of Joint Transformation

The lecture introduced joint transformation which involves:

- Combining two random variables.
- Producing a new random variable using a function of both variables.
- Studying derived distributions based on the joint behavior of the original variables.

3.3 Logical Step-by-Step Reasoning

1. Start with two random variables.
2. Define a function involving both variables.
3. Apply the function to generate a new random variable.
4. Analyze the derived distribution of the new variable.

3.4 Result Emphasized in Lecture

- Functions of multiple random variables lead to derived distributions.
- Joint transformation extends the idea of transformation from one variable to multiple variables.

4. Illustrative Example: $Z = X + Y$

4.1 Statement of the Example

The lecture provided a detailed derivation example using:

$$Z = X + Y$$

4.2 Objective of the Example

- To demonstrate practical application of joint transformation.
- To illustrate how a new random variable is formed from two variables.

4.3 Step-by-Step Reasoning Provided

1. Consider two random variables **X** and **Y**.
2. Define a new variable **Z** as the sum of the two variables.
3. The value of **Z** depends on the combined values of **X** and **Y**.
4. The derived distribution of **Z** is determined from the joint behavior of **X** and **Y**.

4.4 Key Insight from Example

- Summation of two random variables is a special case of transformation involving two variables.

- The example demonstrates how transformation techniques are applied to derive new distributions.

5. Logical Flow of Ideas Across the Lecture

The lecture followed the progression:

1. Introduction to transformation of a single random variable.
2. Extension of transformation techniques to functions involving two random variables.
3. Application of concepts using the summation example.

6. Key Takeaways

- Transformation methods allow creation and analysis of new random variables.
- Joint transformations consider relationships between multiple variables.
- Derived distributions describe outcomes resulting from transformations.
- The example $Z = X + Y$ demonstrates practical implementation of transformation concepts.

(End of Lecture 11 Scribe – Strictly reconstructed from provided PPT content.)

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Lecture Scribe – Lecture 11

Course: CSE400 – Fundamentals of Probability in Computing

Lecture: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Date: February 10, 2026

Scribe Preparation Method:

Prepared using **self-consistency prompting** with identical prompt and context. The reconstruction strictly follows PPT content only, ensuring consistent logical reconstruction without external additions.

1. Lecture Overview

1.1 Learning Objectives

The lecture focuses on three key areas:

1. Transformation of Random Variables

- Learning transformation techniques for random variables.

2. Function of Two Random Variables

- Understanding joint transformations and derived distributions.

3. Illustrative Example

- Detailed derivation for:

$$Z = X + Y$$

2. Transformation of Random Variables

2.1 Definition and Notation

- Let X represent an original random variable.
- Let Z represent a transformed random variable.
- The transformation is expressed as:

$$Z = g(X)$$

2.2 Assumptions and Conditions

- A transformation is performed using a function applied to an existing random variable.
 - The output of the function is also treated as a random variable.
-

2.3 Conceptual Reasoning and Logical Flow

The lecture develops the concept using the following logical steps:

1. Begin with an existing random variable.
 2. Apply a transformation function to the variable.
 3. The transformation produces a new random variable.
 4. Transformation techniques allow study of the behavior of the new variable.
-

2.4 Result

- Transformation techniques help derive new random variables from existing ones.
 - These techniques help analyze resulting probability distributions.
-

3. Function of Two Random Variables

3.1 Definition and Notation

- Let X and Y denote two random variables.
- Let Z denote a derived random variable defined by:

$$Z = g(X, Y)$$

3.2 Joint Transformation

Joint transformation refers to:

- Applying a function to two random variables.
 - Producing a new random variable using both variables.
 - Studying derived distributions from their joint behavior.
-

3.3 Assumptions and Conditions

- Two random variables are considered simultaneously.
 - A function combining both variables is applied.
 - The resulting output is treated as a derived random variable.
-

3.4 Step-wise Logical Development

1. Identify two random variables X and Y .
 2. Define a transformation function involving both variables.
 3. Generate a new variable using the transformation.
 4. Study the derived distribution of the new variable.
-

3.5 Result

- Joint transformations extend single variable transformations.
 - Derived distributions depend on the combined behavior of both variables.
-

4. Illustrative Example

4.1 Statement

The lecture presents an example:

$$Z = X + Y$$

4.2 Objective of Example

- Demonstrate practical application of transformation of two random variables.
 - Illustrate derivation of a new random variable from two variables.
-

4.3 Step-by-Step Logical Reasoning

Step 1: Consider two random variables X and Y .

Step 2: Define a new random variable:

$$Z = X + Y$$

Step 3: The value of Z depends on the combined values of X and Y .

Step 4: The distribution of Z is derived from the joint behavior of X and Y .

4.4 Result of Example

- Summation of random variables is a special case of joint transformation.
 - Derived distributions arise from applying transformation techniques.
-

5. Overall Logical Flow of Lecture

The lecture follows the progression:

1. Introduces transformation of a single random variable.
 2. Extends transformation to two random variables.
 3. Demonstrates concept using summation example.
-

6. Key Takeaways

- Transformation techniques create new random variables.
- Joint transformation studies functions involving multiple variables.
- Derived distributions describe behavior of transformed variables.
- The example $Z = X + Y$ illustrates joint transformation.

End of Lecture 11 Scribe

(Strictly reconstructed from PPT content with consistent logical structure.)

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Lecture Topic: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Date: February 10, 2026

Scribe Construction Note:

Prepared using **self-consistency prompting framework** with identical prompt and context. The reconstruction strictly follows information available in the lecture PPT only. No external definitions, derivations, or assumptions are added.

1. Lecture Outline and Learning Goals

The lecture covered three primary components:

1.1 Transformation of Random Variables

- Focus on learning transformation techniques for random variables.

1.2 Function of Two Random Variables

- Study of joint transformations.
- Understanding derived distributions.

1.3 Illustrative Example

- Detailed derivation for:

$$Z = X + Y$$

2. Transformation of Random Variables

2.1 Definitions and Notation

- X : Original random variable.
- Z : Random variable obtained after transformation.

Transformation is expressed as:

$$Z = g(X)$$

2.2 Assumptions and Conditions

- A mathematical function is applied to an existing random variable.
 - The output produced by the function is considered another random variable.
-

2.3 Logical Development of the Concept

Step 1: Begin with a known random variable X .

Step 2: Apply a transformation function $g(\cdot)$ to X .

Step 3: The transformation produces a new random variable Z .

Step 4: Transformation techniques are used to study the characteristics and behavior of Z .

2.4 Result

- Transformation methods allow derivation of new random variables.

- These methods enable analysis of distributions obtained after transformation.

3. Function of Two Random Variables

3.1 Definitions and Notation

- X, Y : Two random variables.
- Z : Derived random variable defined by a function of X and Y .

General representation:

$$Z = g(X, Y)$$

3.2 Joint Transformation

Joint transformation involves:

- Applying a function to two random variables simultaneously.
- Generating a new random variable from both.
- Studying distributions derived from their joint behavior.

3.3 Assumptions and Conditions

- Two random variables are considered together.
- A transformation function combines both variables.
- The resulting value is treated as a new random variable.

3.4 Step-by-Step Logical Flow

Step 1: Identify two random variables X and Y .

Step 2: Define a transformation function involving both variables.

Step 3: Apply the function to generate a new variable Z .

Step 4: Analyze the derived distribution of Z .

3.5 Result

- Joint transformation extends single variable transformation.
- Derived distributions depend on the combined behavior of the involved variables.

4. Illustrative Example

4.1 Statement

The lecture presents the example:

$$Z = X + Y$$

4.2 Objective of the Example

- To demonstrate transformation involving two random variables.
- To illustrate derivation of a new random variable through summation.

4.3 Step-by-Step Development

Step 1: Consider two random variables X and Y .

Step 2: Define the new random variable:

$$Z = X + Y$$

Step 3: The value of Z depends on combined values of X and Y .

Step 4: The distribution of Z is derived from the joint behavior of X and Y .

4.4 Result

- The summation of two random variables represents a special case of joint transformation.
- The example illustrates how derived distributions are formed.

5. Overall Logical Flow of the Lecture

The lecture progresses in the following structured sequence:

1. Introduces transformation of a single random variable.
2. Extends the transformation concept to two random variables.
3. Demonstrates the concept using an example involving summation.

6. Key Conclusions

- Transformation techniques allow creation of new random variables.
- Joint transformations involve functions of multiple variables.
- Derived distributions describe outcomes of transformations.
- The example $Z = X + Y$ provides a direct application of joint transformation.

End of Lecture 11 Scribe

(Strict reconstruction based solely on lecture PPT content.)